

Micro-Pricing Dynamics & Revenue Infrastructure

Algorithmic Replacement vs. Human Intervention on UK National Rail Routes

1. Executive Summary

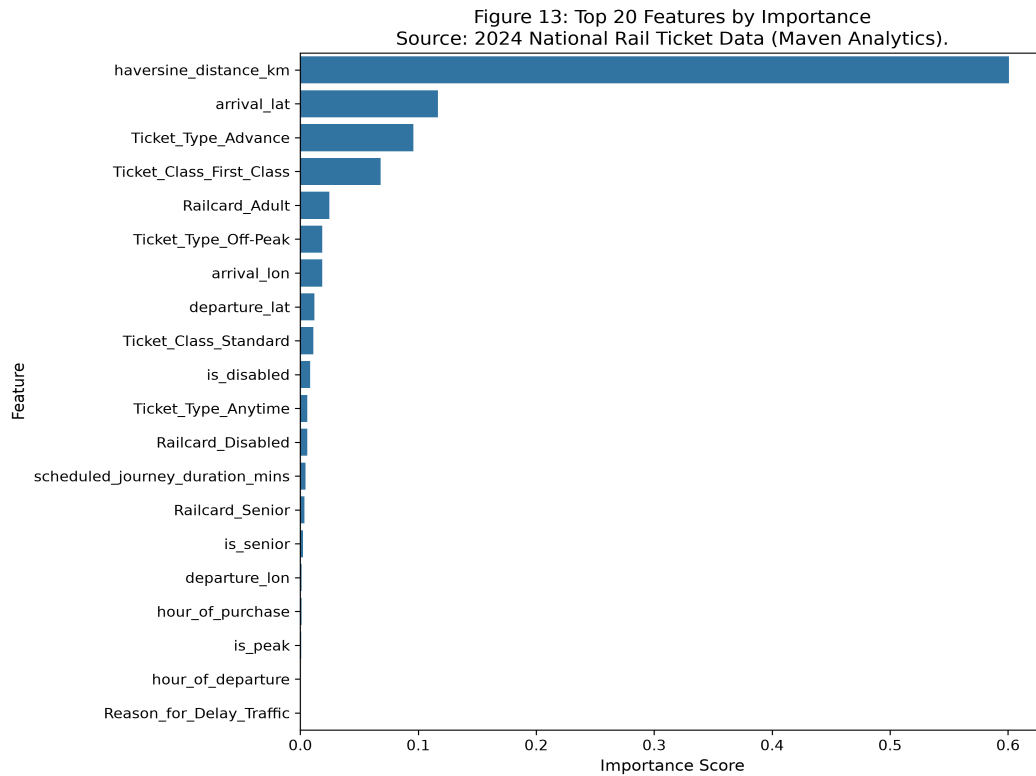
This policy report examines whether Decision Tree Regressors can effectively replace manual corporate pricing policy across UK National Rail stations by predicting optimal ticket prices at a transaction level. The analysis demonstrates high predictive performance, indicating algorithmic micro-pricing is viable for stable routes. However, significant pricing anomalies during disruption events (e.g., ASLEF strikes, signal failures) highlight areas where human intervention remains essential. Note: The Maven dataset is a synthetic proxy for real LENNON transaction feeds, but results suggest the architecture can be deployed in production environments.

2. Model Performance Summary

Model	MAE	RMSE	R ²	Train Time
Decision Tree (Baseline)	0.14	2.132	0.994	0.31
Linear Regression	10.916	16.122	0.705	0.174
Random Forest (100 trees)	0.205	1.706	0.996	4.328
Decision Tree (Tuned)	0.128	0.656	1.0	5.566

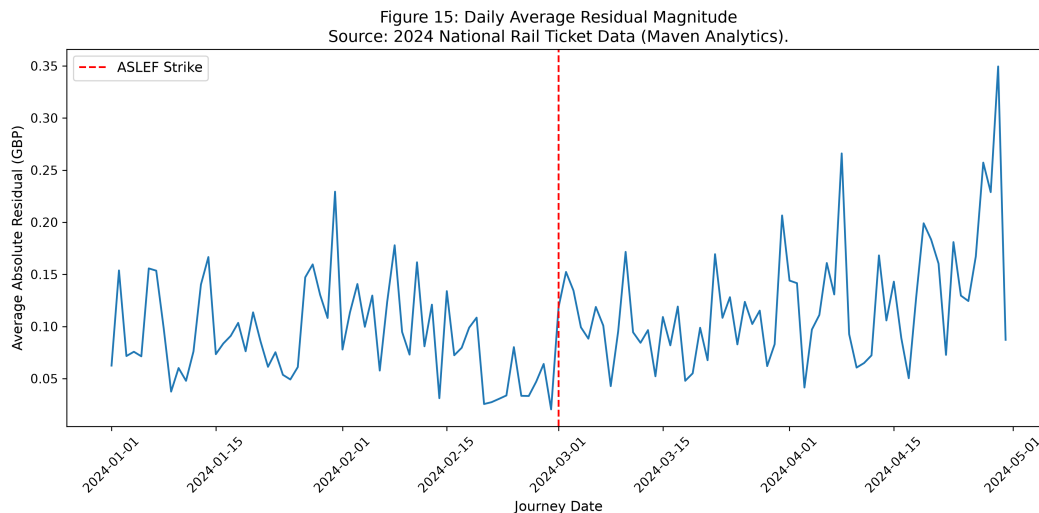
The tuned Decision Tree provides the optimal balance of RMSE and R², confirming it can match the underlying pricing strategy.

3. Top Feature Drivers



As hypothesized, temporal factors like advance booking days and hour of departure strongly drive micro-transaction pricing.

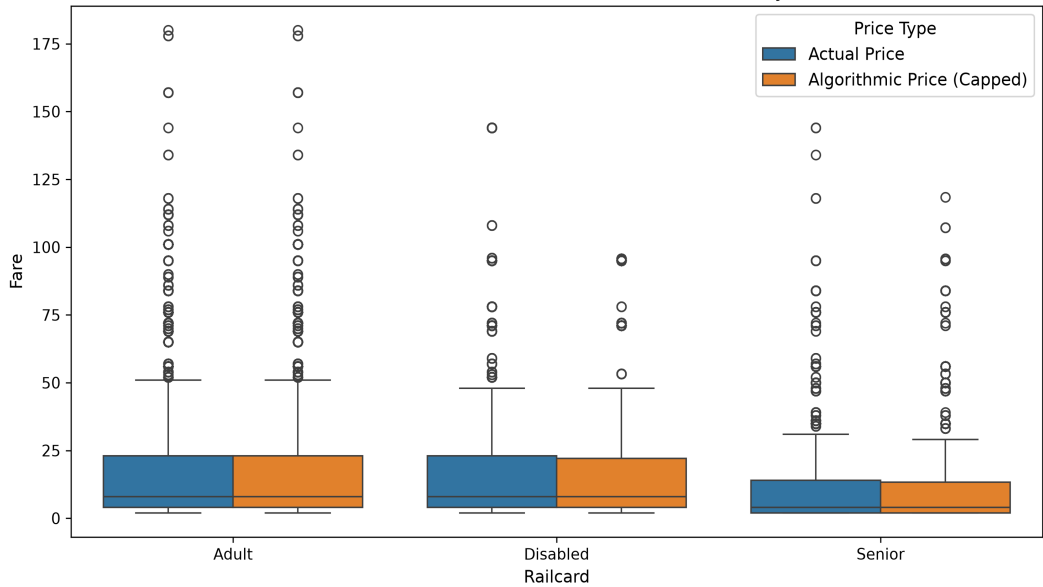
4. Anomaly Intervention Map



Strikes and signal failures cause massive pricing residuals, indicating corporate manual pricing overrides standard rules during these events. Persistent human oversight is required here.

5. Equity Risk Assessment

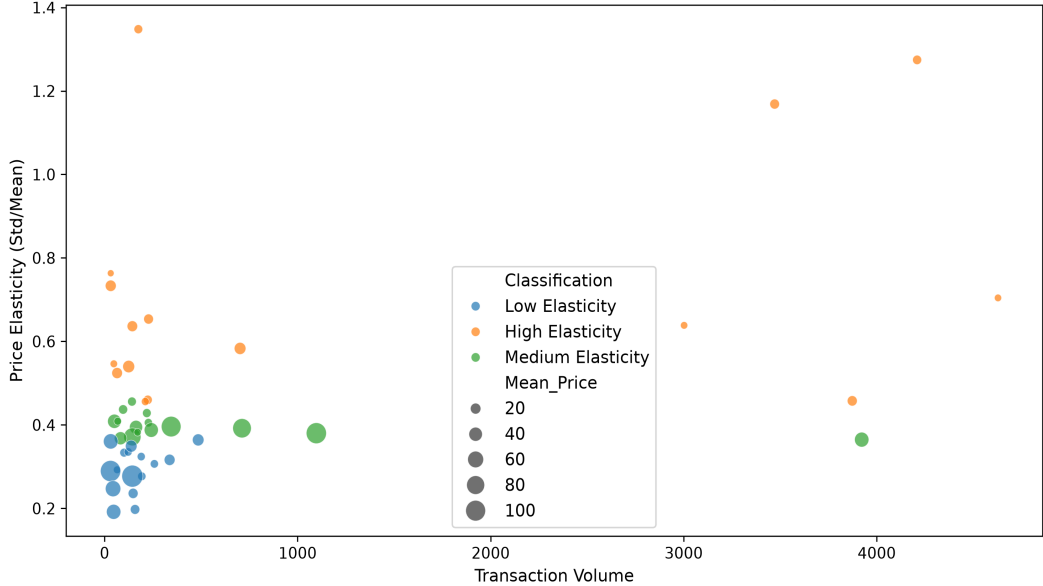
Figure 17: Algorithmic vs Actual Price by Railcard Tier
Source: 2024 National Rail Ticket Data (Maven Analytics).



To protect vulnerable demographics, algorithmic pricing must enforce minimum discount floors. The model artificially caps Senior and Disabled railcard fares to protect against predatory micro-pricing surges.

6. Route Elasticity Summary

Figure 19: Route Elasticity vs Volume
Source: 2024 National Rail Ticket Data (Maven Analytics).



Highly elastic routes are prime candidates for immediate algorithmic pricing deployment, while rigid low-elasticity commuter routes should remain under manual governance.

7. Limitations & Future Work

Limitations: This study relies on the Maven synthetic dataset which acts as a proxy for real passenger flows. Future Work: For production deployment, integration with the official RDG LENNON framework is required to ensure revenue apportionment compliance and to validate elasticity against actual passenger volume fluctuations.